The Enhanced Dold-Kan Correspondence

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Table of Contents

Simplicial sets

The Dold-Kan correspondence

Augmented simplicial abelian groups

The enhanced Dold-Kan correspondence

Extensions

Appendix

Simplicial sets

Let $\mathbb{\Delta}$ be the category of finite, totally ordered sets, together with order-preserving maps. Write objects in $\mathbb{\Delta}$ as $[n] = \{0 \dots n\}$ with the usual order.

Definition

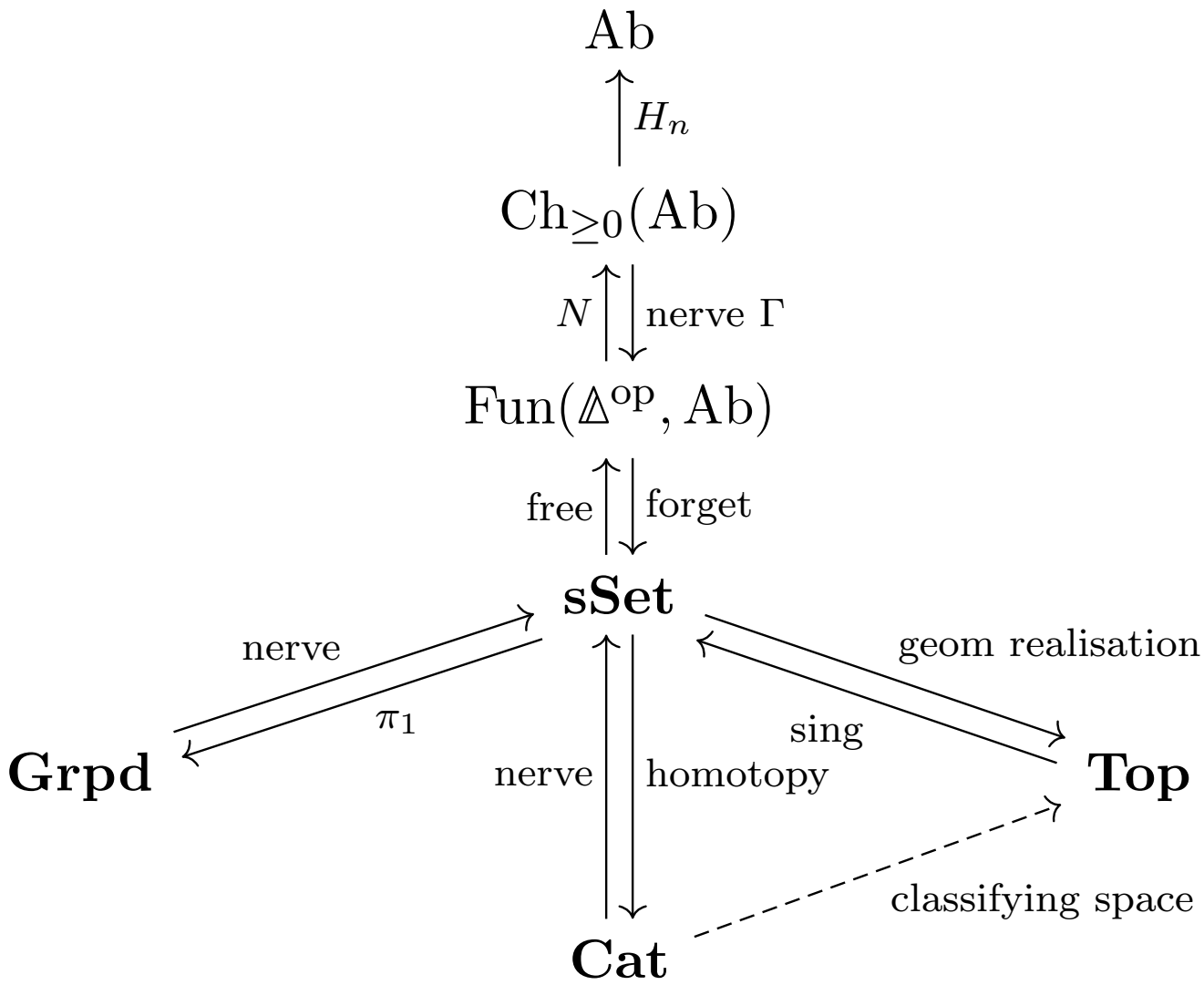
A simplicial abelian group is a functor

$$A_\bullet \in \mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathrm{Ab})$$

written $A_\bullet = (A_n)_{n \geq 0}$, where $A_n \in \mathcal{C}$ is the image of $[n]$.

Note that a simplicial set $X_\bullet \in \mathbf{sSet}$ is a functor $\mathbb{\Delta}^{\mathrm{op}} \longrightarrow \mathbf{Set}$.

Simplicial sets are unifying



Each pair of arrows is an adjunction.

What is the functor $N : \mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathrm{Ab}) \longrightarrow \mathrm{Ch}_{\geq 0}(\mathrm{Ab})$?

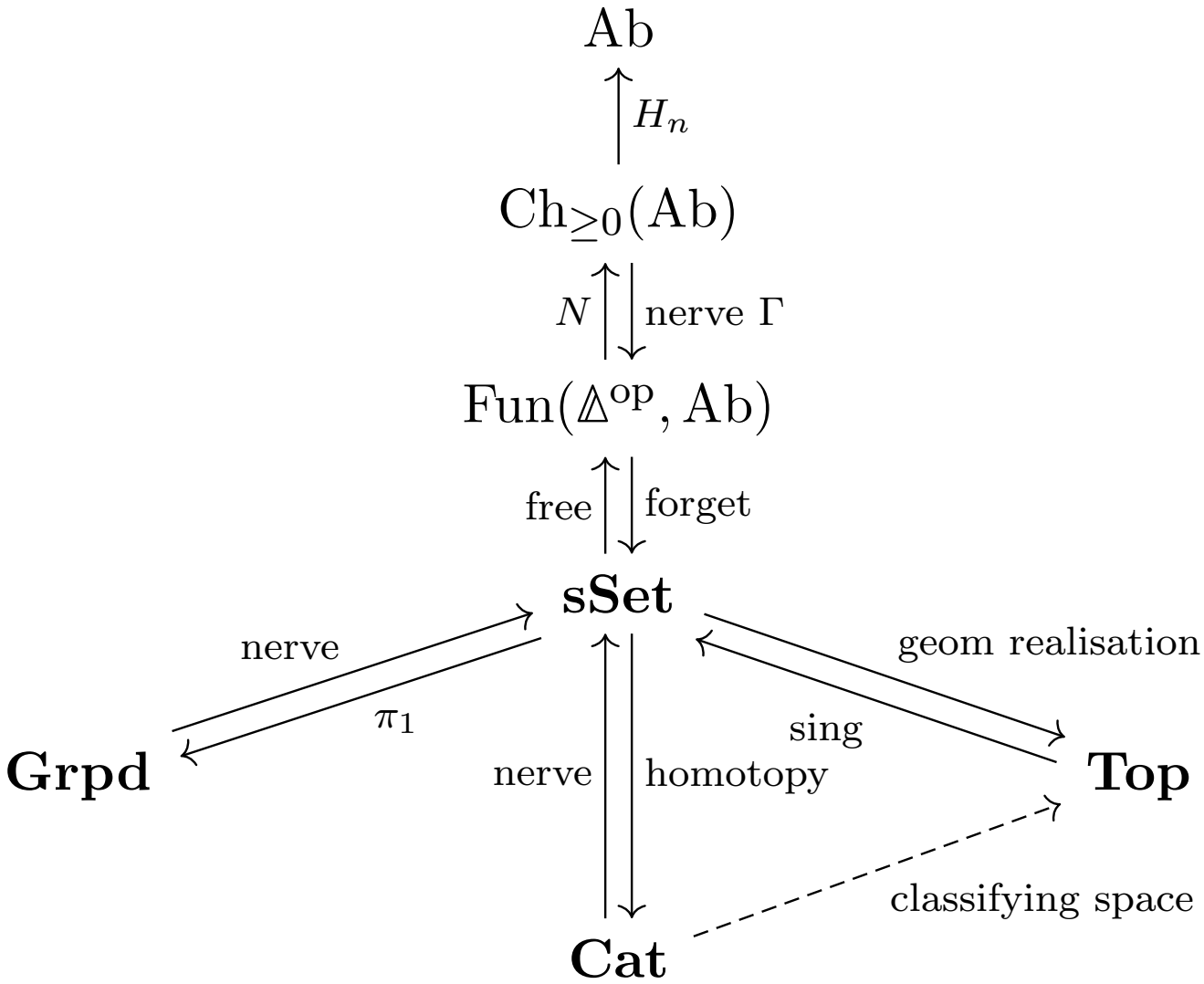


Table of Contents

Simplicial sets

The Dold-Kan correspondence

Augmented simplicial abelian groups

The enhanced Dold-Kan correspondence

Extensions

Appendix

Face maps

In $\mathbb{\Delta}$ there are special maps. For $n \geq 0$ and $0 \leq i \leq n$

$$d_n^i : [n-1] \longrightarrow [n]$$
$$(0 \to \cdots \to n-1) \mapsto (0 \to \cdots \to i-1 \to i+1 \to \cdots \to n)$$

i.e. compose $i-1 \to i \to i+1$.

Under the functor $A_\bullet : \mathbb{\Delta}^{\mathrm{op}} \longrightarrow \mathrm{Ab}$ we call this map

$$d_i^n : A_n \longrightarrow A_{n-1}$$

a *face* map.

The normalised complex

Let

$$NA_n = \bigcap_{i<n} \ker(d_i^n : A_n \longrightarrow A_{n-1}) \subseteq A_n$$

for $n \geq 1$ and $NA_0 = A_0$.

Lemma

$$(NA_n, (-1)^n d_n^n : NA_n \longrightarrow NA_{n-1})_{n\geq 0}$$

is a chain complex, called the normalised chain complex $NA_\bullet$.

The classical Dold-Kan correspondence

N is a functor $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Ab}) \longrightarrow \mathrm{Ch}_{\geq 0}(\mathrm{Ab})$. Moreover,

Theorem (Dold, 1958 and Kan, 1958)

$$N : \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathrm{Ab}) \longrightarrow \mathrm{Ch}_{\geq 0}(\mathrm{Ab})$$

is an equivalence of categories.

Historical Note

The Dold-Kan correspondence was proven at similar times by Dold and Kan, both publishing in 1958.

The result was extended to all abelian categories by Dold and Puppe 1961.

Kan's article also introduced *Kan complexes*.

Table of Contents

Simplicial sets

The Dold-Kan correspondence

Augmented simplicial abelian groups

The enhanced Dold-Kan correspondence

Extensions

Appendix

Monidal structure in Δ?

Note: Δ does not have a tensor product—the category is *not monoidal*.

One can consider Δ_η: Δ with an initial object $[-1]$. Δ_η is called the *augmented simplex category*.

Lemma

$(\Delta_\eta, \sqcup, \emptyset)$ is a strict monoidal category, where

$$[n] \sqcup [m] := [n + m + 1].$$

Algebra objects

Definition

An monoid object A in $(M, \otimes, 1)$ is an object with arrows

$$m : A \otimes A \to A, \quad \eta : 1 \to A.$$

(the *multiplication* and *unit* arrows).

In Δ_η there are arrows

$$[1] = [0] \sqcup [0] \longrightarrow [0], \quad [-1] \longrightarrow [0].$$

Hence $[0]$ is the monoid object in Δ_η.

Δ_η classifies monoid objects

There are many reasons to work with Δ_η over Δ. One is due to MacLane.

Theorem (MacLane, 2013)

Let A be a monoid in a strict monoidal category $(M, \otimes, 1)$. There a unique arrow with diagram

$$\begin{array}{ccccccc} \Delta_\eta & & [1] = [0] \sqcup [0] & \longrightarrow & [0] & \longleftarrow & [-1] \\ {\scriptstyle \exists! F}\,\vdots\downarrow & & \vdots\downarrow & & \vdots\downarrow & & \vdots\downarrow \\ M & & A \otimes A & \longrightarrow & A & \longleftarrow & 1 \end{array}$$

Meaning: Δ_η is the universal, strict monoidal category with a monoid object.

Augmented simplicial abelian groups

An augmented simplicial abelian group is a functor

$$A_\bullet \in \mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathrm{Ab}).$$

Compared to a simplicial abelian group, it comes with an extra map $\epsilon : A_0 \longrightarrow A_{-1}$.

$$A_{-1} \xleftarrow{\epsilon} A_0 \underset{d_1^1}{\overset{d_0^1}{\leftleftarrows}} A_1 \;\substack{\leftarrow\\ \leftarrow\\ \leftarrow}\; A_2 \;\substack{\leftarrow\\ \leftarrow\\ \leftarrow\\ \leftarrow}\; \dots$$

ϵ coequalises the face maps $\epsilon d_0^1 = \epsilon d_1^1$.

Augmented normalised complex

One can construct an analogue to N, called N^a. Given $A_\bullet \in \mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathrm{Ab})$ it produces the chain complex

$$A_{-1} \xleftarrow{\epsilon} A_0 \xleftarrow{(-1)d_1^1} NA_1 \xleftarrow{(-1)^2 d_2^2} NA_2 \longleftarrow \dots$$

$$\cdots \longleftarrow NA_{n-1} \xleftarrow{(-1)^n d_n^n} NA_n \longleftarrow \dots$$

This is called an *augmented chain complex* (i.e. we add a term in degree -1).

Question: Is $N^a : \mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathrm{Ab}) \longrightarrow \mathrm{Ch}_{\geq -1}(\mathrm{Ab})$ an equivalence of categories?

Table of Contents

Simplicial sets

The Dold-Kan correspondence

Augmented simplicial abelian groups

The enhanced Dold-Kan correspondence

Extensions

Appendix

Main theorem

Theorem (The enhanced Dold-Kan correspondence)

$$N^a : \mathrm{Fun}({\Delta_\eta}^{\mathrm{op}}, \mathrm{Ab}) \longrightarrow \mathrm{Ch}_{\geq -1}(\mathrm{Ab})$$

is an equivalence of categories.

The proof: comma categories are the lingua franca

A natural way to express the extra strcture in

$$\mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathrm{Ab}) \quad \text{and} \quad \mathrm{Ch}_{\geq -1}(\mathrm{Ab})$$

compared to

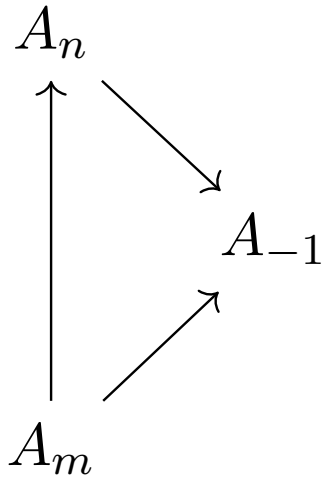
$$\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathrm{Ab}) \quad \text{and} \quad \mathrm{Ch}_{\geq 0}(\mathrm{Ab})$$

is via *comma categories*.

Identification 1

$A_\bullet = (A_n)_{n \geq -1} \in \mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathrm{Ab})$ is

$(A_n)_{n \geq 0} \in \mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathrm{Ab})$ with



$\forall n, m \geq 0$

for some $A_{-1} \in \mathrm{Ab}$.

Dold-Kan

Under Dold-Kan we identify the above data with

$$N(A_n)_{n\geq 0} \in \mathrm{Ch}_{\geq 0}(\mathrm{Ab}) \qquad \text{and} \qquad \begin{array}{c}\text{chain map}\\ N(A_n)_{\geq 0} \longrightarrow A_{-1}[0]\end{array}$$

$$\begin{array}{ccccccccc}
A_0 & \xleftarrow{(-1)d_1^1} & NA_1 & \xleftarrow{(-1)^2 d_2^2} & NA_2 & \xleftarrow{(-1)^3 d_3^3} & \dots & = (A_n)_{\geq 0} \\
\downarrow{\scriptstyle \epsilon} & & \downarrow & & \downarrow & & & \downarrow \\
A_{-1} & \longleftarrow & 0 & \longleftarrow & 0 & \longleftarrow & \dots & = A_{-1}[0]
\end{array}$$

Identification 2

Which we can identify with a chain complex

$$N^a(A_n)_{n\geq -1} = N^a A_\bullet \in \mathrm{Ch}_{\geq -1}(\mathrm{Ab})$$

$$A_{-1} \xleftarrow{\epsilon} A_0 \xleftarrow{(-1)d_1^1} A_1 \xleftarrow{(-1)^2 d_2^2} A_2 \longleftarrow \cdots = N^a A_\bullet$$

These identifications are all functorial and equivalences involving comma categories. They compose to give N^a.

We have proven

Lemma (Identification 1)

$$\mathrm{Fun}(\mathbb{\Delta}_{\eta}{}^{\mathrm{op}}, \mathrm{Ab}) \simeq (\mathrm{Fun}(\mathbb{\Delta}, \mathrm{Ab}) \downarrow \mathrm{Ab}).$$

Lemma (Dold-Kan on comma categories)

$$(\mathrm{Fun}(\mathbb{\Delta}, \mathrm{Ab}) \downarrow \mathrm{Ab}) \simeq (\mathrm{Ch}_{\geq 0}(\mathrm{Ab}) \downarrow \mathrm{Ab}).$$

Lemma (Identification 2)

$$(\mathrm{Ch}_{\geq 0}(\mathrm{Ab}) \downarrow \mathrm{Ab}) \simeq \mathrm{Ch}_{\geq -1}(\mathrm{Ab}).$$

One can write down, explicitly, what the functors going left-right are.

Table of Contents

Simplicial sets

The Dold-Kan correspondence

Augmented simplicial abelian groups

The enhanced Dold-Kan correspondence

Extensions

Appendix

Extensions: Abelian categories

Replace Ab with an (any?) abelian category $\mathcal{A}$. Then

Conjecture: $N^a : \mathrm{Fun}(\Delta_\eta{}^{\mathrm{op}}, \mathcal{A}) \xrightarrow{\simeq} \mathrm{Ch}_{\geq -1}(\mathcal{A})$.

(Be careful—abelain categories are not always *concrete*!)

Extensions: Quillen equivalence

Theorem (Quillen, 1967)

The classical Dold-Kan correspondence extends to a Quillen equivalence.

Among other things, this means N induces an equivalence of the homotopy categories in the Dold-Kan correspondence.

This simplifies doing homotopy theory: one passes, without loss of information, from combinatorial objects (simplicial abelian groups) to algebraic objects (chain complexes).

Extensions—Quillen equivalence

Conjecture: N^a extends to a Quillen equivalence.

There is work to do here—one needs the theory of *model categories*.

Table of Contents

Simplicial sets

The Dold-Kan correspondence

Augmented simplicial abelian groups

The enhanced Dold-Kan correspondence

Extensions

Appendix

Homotopy is not concrete

Freyd, P. (2019). Homotopy is not concrete.
https://www2.math.upenn.edu/~pjf/homotopy.pdf

What is a monoidal category?

Definition

A *monoidal* category M is a category, together with the following.

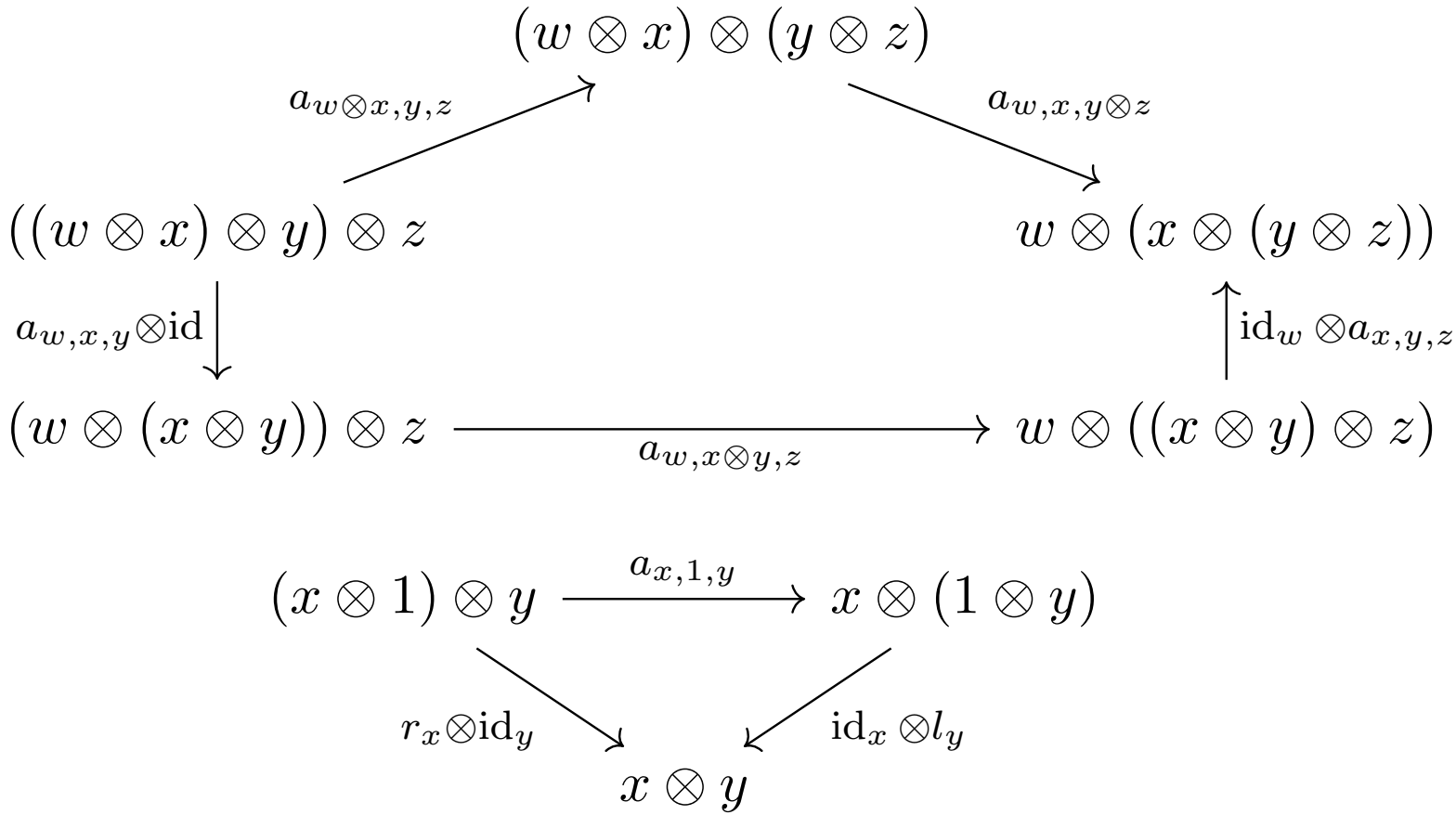
1. a *bifunctor* $\otimes : M \times M \to M$ (write $x \otimes y$ and $f \otimes g$ for the image of two objects and morphisms under $\otimes$) called the *tensor product*.
2. an identity object $1 \in M$.
3. Natural isomorphisms

$$a_{x,y,z} : (x \otimes y) \otimes z \xrightarrow{\sim} x \otimes (y \otimes z)$$

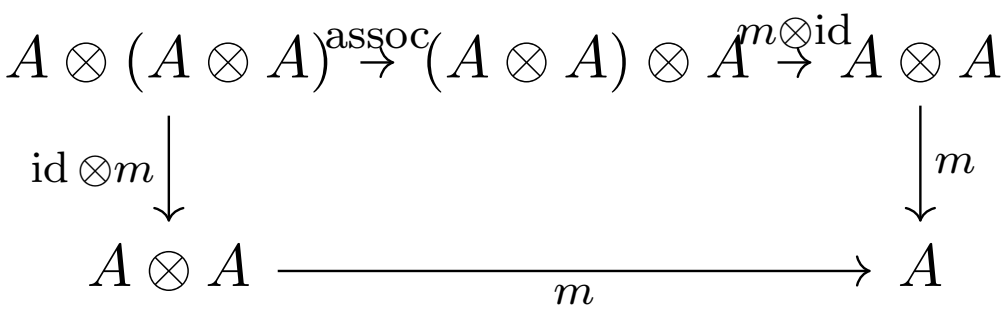
and

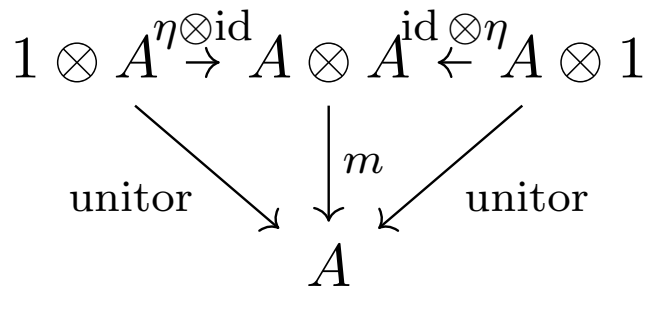
$$l_x : 1 \otimes x \xrightarrow{\sim} x, \quad r_x : x \otimes 1 \xrightarrow{\sim} x.$$

for objects $x, y, z \in M$. These natural isomorphisms are *coherent*.


$(w \otimes x) \otimes (y \otimes z)$
$a_{w\otimes x,y,z}$
$a_{w,x,y\otimes z}$
$((w \otimes x) \otimes y) \otimes z$
$w \otimes (x \otimes (y \otimes z))$
$a_{w,x,y}\otimes \mathrm{id}$
$\mathrm{id}_w \otimes a_{x,y,z}$
$(w \otimes (x \otimes y)) \otimes z$
$w \otimes ((x \otimes y) \otimes z)$
$a_{w,x\otimes y,z}$
$(x \otimes 1) \otimes y$
$a_{x,1,y}$
$x \otimes (1 \otimes y)$
$r_x \otimes \mathrm{id}_y$
$\mathrm{id}_x \otimes l_y$
$x \otimes y$

Coherence conditions for a monoid





Eilenberg-MacLane normalisation theorem

The *Moore* complex of a simplicial abelian group $A_\bullet$ is

$$A_0 \xleftarrow{\partial_1} A_1 \xleftarrow{\partial_2} A_2 \longleftarrow \dots$$

where $\partial_n = \sum_{i \leq n} (-1)^i d_i^n$.

There is an inclusion $NA_\bullet \longrightarrow A_\bullet$

$$\begin{array}{ccccccccc}
\dots & \longleftarrow & NA_{n-1} & \xleftarrow{(-1)^n d_n} & NA_n & \xleftarrow{(-1)^{n+1} d_{n+1}} & NA_{n+1} & \longleftarrow & \dots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
\dots & \longleftarrow & A_{n-1} & \xleftarrow[\partial_n]{} & A_n & \xleftarrow[\partial_{n+1}]{} & A_{n+1} & \longleftarrow & \dots
\end{array}$$

Eilenberg-MacLane normalisation theorem

Theorem (Goerss and Jardine, 2012)

The inclusion $NA_\bullet \longrightarrow A_\bullet$ is a chain homotopy equivalence. In particular,

$$H_n(NA) \cong H_n(A).$$

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